Survivability in Pokémon: An Analysis of HP + Defense + Sp. Defense

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Research Question

By analyzing survivability across various Pokémon, how do trade-offs emerge between stats, and what patterns emerge across generations, types, and legendary status?

Executive Summary

This project examines Pokémon survivability, exploring how it varies across generations, types, and Legendary status. By analyzing a dataset of 1,025 Pokémon, we calculated a survivability index to quantify a Pokémon's defensive strength through the formula: HP + Defense + Special Defense. Through data cleaning and visual analysis, we examined patterns in survivability, identified potential trade-offs with offensive stats, and explored power creep across generations. Some of our findings include: legendary Pokémon typically exhibit higher survivability, certain types, when combined, produce more survivable Pokémon, and despite possible ideas, defensive index does not negatively correlate with offensive stats.

Methods

Dataset. We used the Kaggle dataset of 1,025 Pokémons for our analysis. Columns included names, types, stats (HP, Attack, Defense, Sp. Atk, Sp. Def, Speed, Total), generation, and legendary status.

Cleaning. Duplicate rows were identified and removed, and those with missing values were removed.

Feature Engineering. We created three new features:

- **Survivability Index:** HP + Defense + Special Defense.
    - This summarizes the defensive capabilities of a Pokémon, and was the primary focus of our analysis

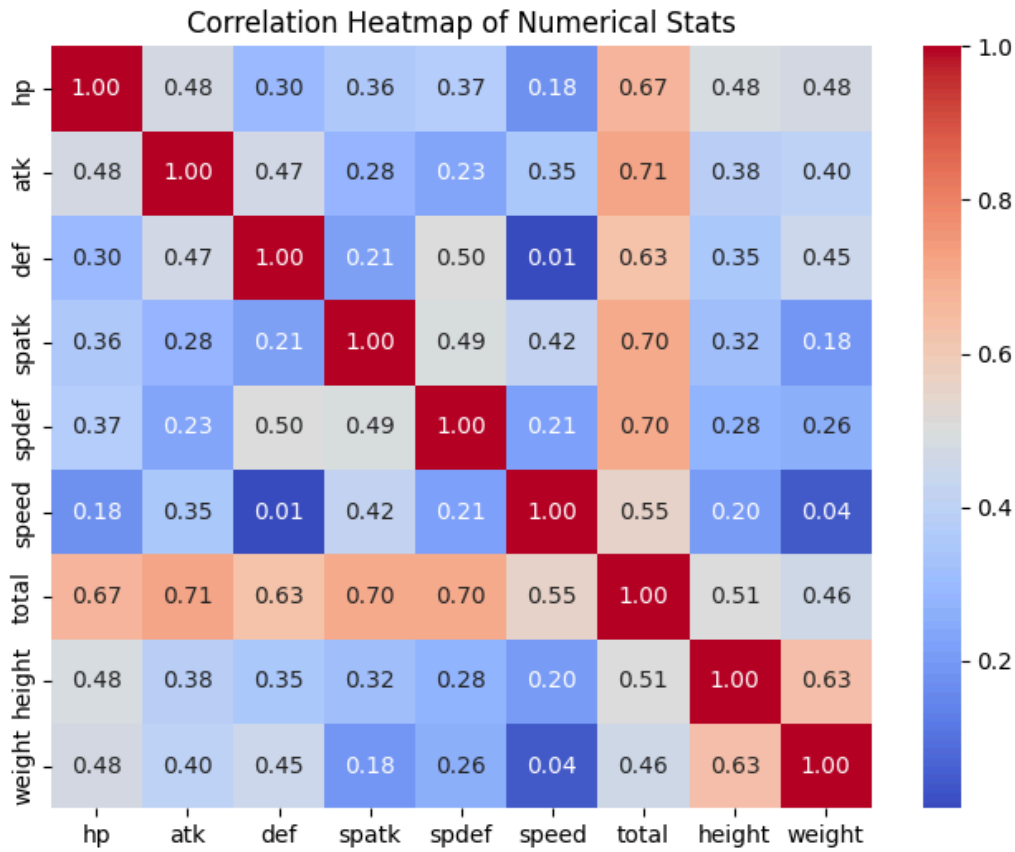
- **Offensive Index:** Attack + Sp. Atk

    - This summarizes the offensive capabilities of a Pokémon.

- **Balance Index:** Standard Deviation of the Columns
    - This is used to determine how balanced or specialized a Pokémon is.
    - Higher values mean the stats are spread apart; therefore, a Pokémon is specialized in one stat
    - Lower values mean the stats are closer together; therefore, a Pokémon is balanced.

General Findings

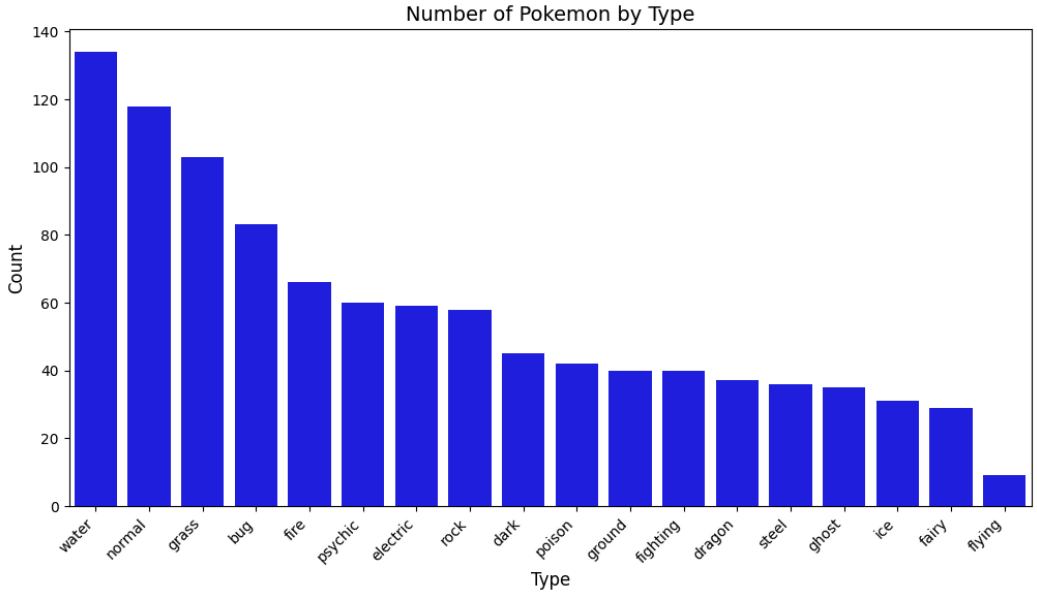
Correlation Heatmap



When looking at the heatmap, which visualizes how each numerical stat is correlated, you can see that there are no negative values.

Therefore, you can determine that any increase in a stat will lead to an increase in every other stat, and vice versa.

Number of Pokémon by Type

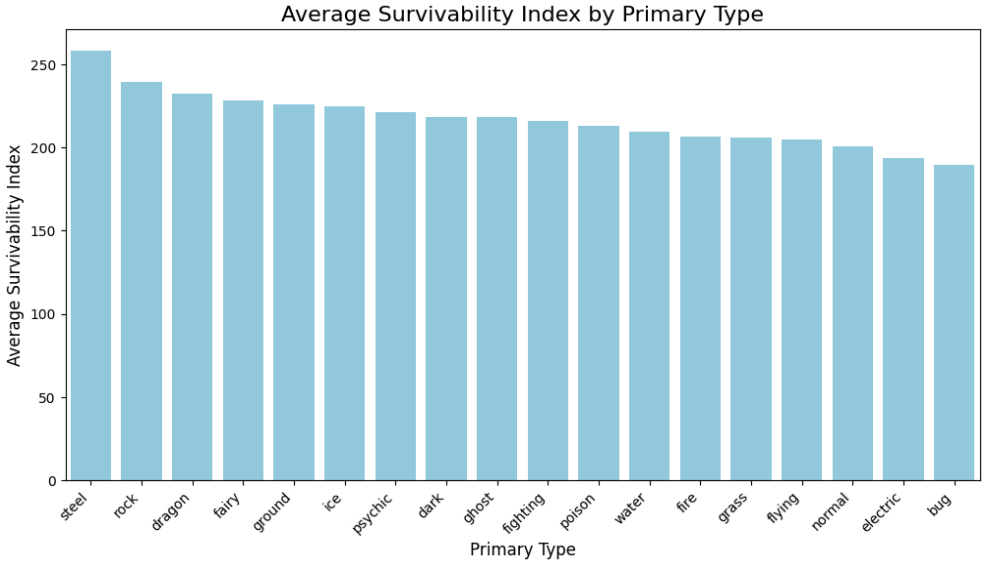


This bar graph shows the most common types of Pokémon in the dataset, allowing us to make informed decisions about future findings related to type count.

The most common primary types were water, normal, and grass, with fairy and steel less common but often defensively specialized.

Survivability Findings

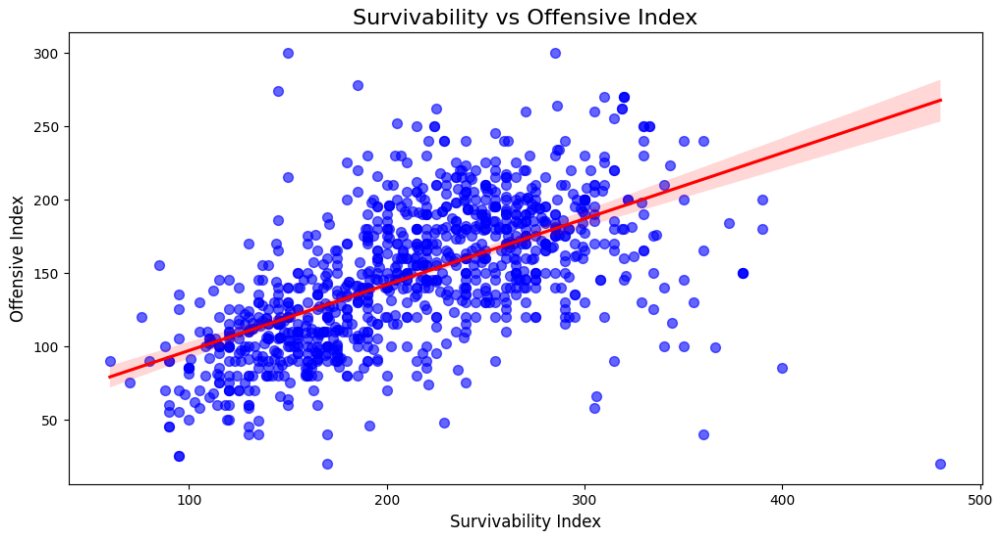
Average Survivability Index by Type



The average survivability index was relatively similar across the types; however, the largest was the steel type, and the smallest was the bug type.

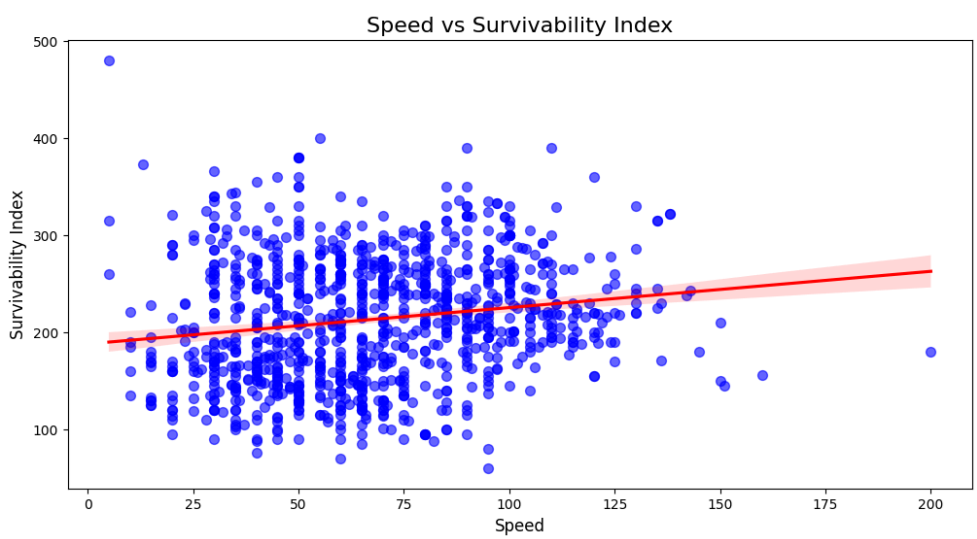
Many intuitively “durable” types have the highest survivability index, such as steel, rock, dragon, ground, etc.

Survivability vs Offensive Index



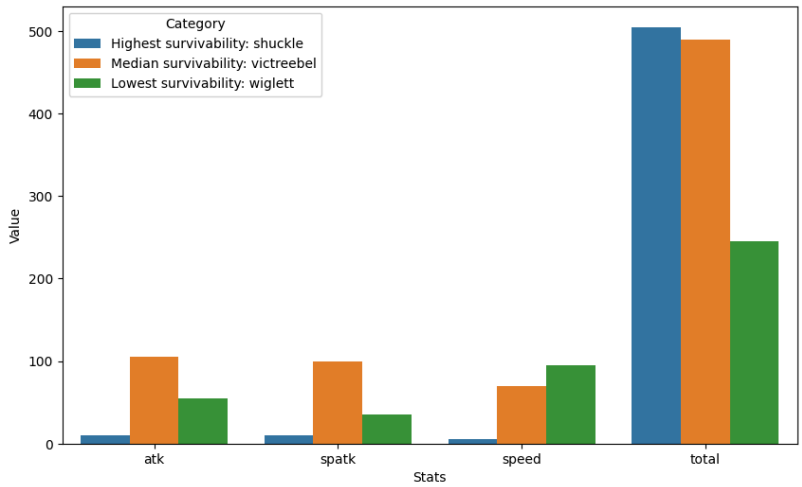
This scatterplot compares Pokémon survivability index to their offensive index (Attack + Special Attack). A red trend line is included in the scatterplot, and based on this, we can conclude that there is no significant trade-off between offense and defense, as increasing survivability also increases offensive ability.

Speed vs Survivability Index



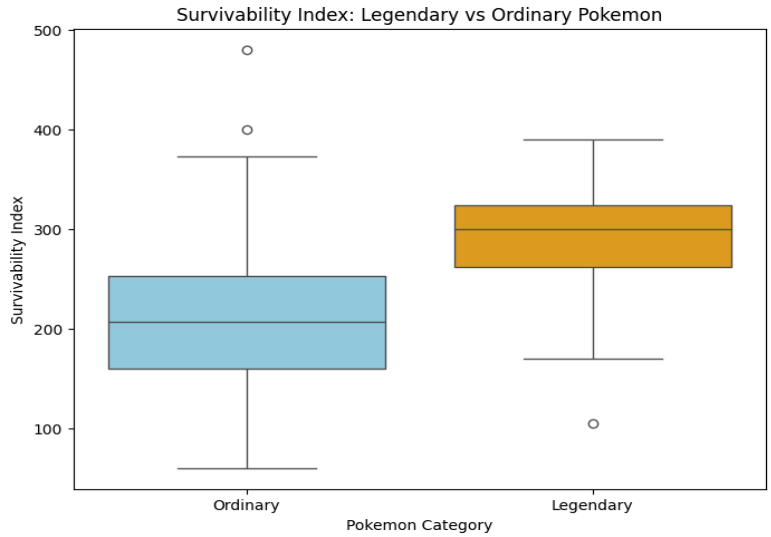
This scatterplot compares speed vs survivability index, and helps to visualize how they are correlated. Despite the intuition that survivability (generally associated with larger size because it could increase HP and Def) may decrease speed, it instead increases speed, though at a low rate.

Stats of Individual Pokémon



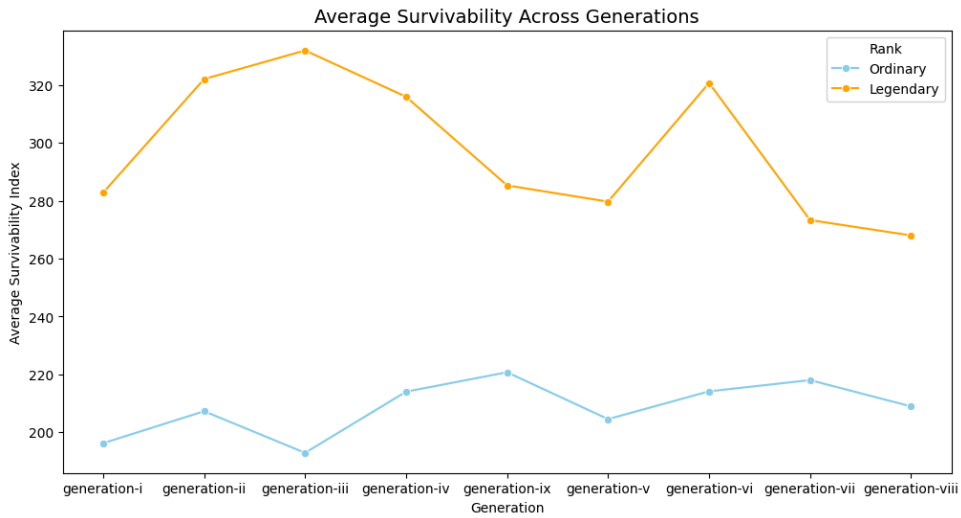
This grouped bar graph allows us to see the stats of the most survivable, least survivable, and median survivable Pokémon. It allows us to visualize the tradeoffs between survivability and total power. Our findings suggest that having high survivability will generally limit other stats, except for total power, which is increased heavily by the survivability. Median survivability leads to the most balanced stats as it has high values in all of its stats and high total power. Low survivability leads to minimal power, as the Pokémon does not compensate for low survivability with extremely high values in its other stats.

Legendary vs. Non-Legendary



On average, Legendaries scored higher in survivability, outmatched by only a few outliers in the ordinary category.

Power Creep Analysis



This graph allows us to see how average survivability has changed across generations for both legendary and ordinary Pokémon.

Across each generation, there have been significant changes in survivability with large spikes and dips. For legendaries, the highest survivability was seen in generation-iii, and for ordinary Pokémon, the highest survivability was seen in generation-ix.

Outlier Analysis

Shuckle epitomizes the tradeoffs between offense and survivability. Shuckle has an absurd survivability index of 480 due to its high Sp. Defense and Defense. Despite its high survivability, the Pokémon is limited by its low offense, with an Attack and Sp. Attack of 10. Shuckle's case suggests that high survivability does not guarantee capability in battle because while Shuckle can survive many attacks, it is difficult for it to deal a decisive blow to its opponent.

Limitations & Future Directions

Our survivability index treats HP, Sp. Defense, and Defense is considered equally valuable in all situations, which is unrealistic in actual gameplay.

In the future, we could weight each variable differently based on factors surrounding a Pokémon's other stats.

In the future, we could incorporate battle simulations to determine how survivability truly affects the effectiveness of a Pokémon.

References

- Kaggle. *Data of 1,025 Pokémons*